

Anytime-Constrained Equilibria in Polynomial Time

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Problem

An anytime-constrained equilibrium (ACE) is a policy π satisfying, for all players i and deviation policies π'_i ,

$$\mathbb{P}_G^{(\pi'_i, \pi_{-i})} \left[\exists t, \sum_{h=1}^t c_h > B \right] > 0 \quad \text{OR} \quad V_i^{(\pi'_i, \pi_{-i})} \leq V_i^\pi$$

Such constraints can model modern many safety and resource consumption requirements.

Challenges

- Computation is **NP-hard**
- Determining **existence** is NP-hard

Results

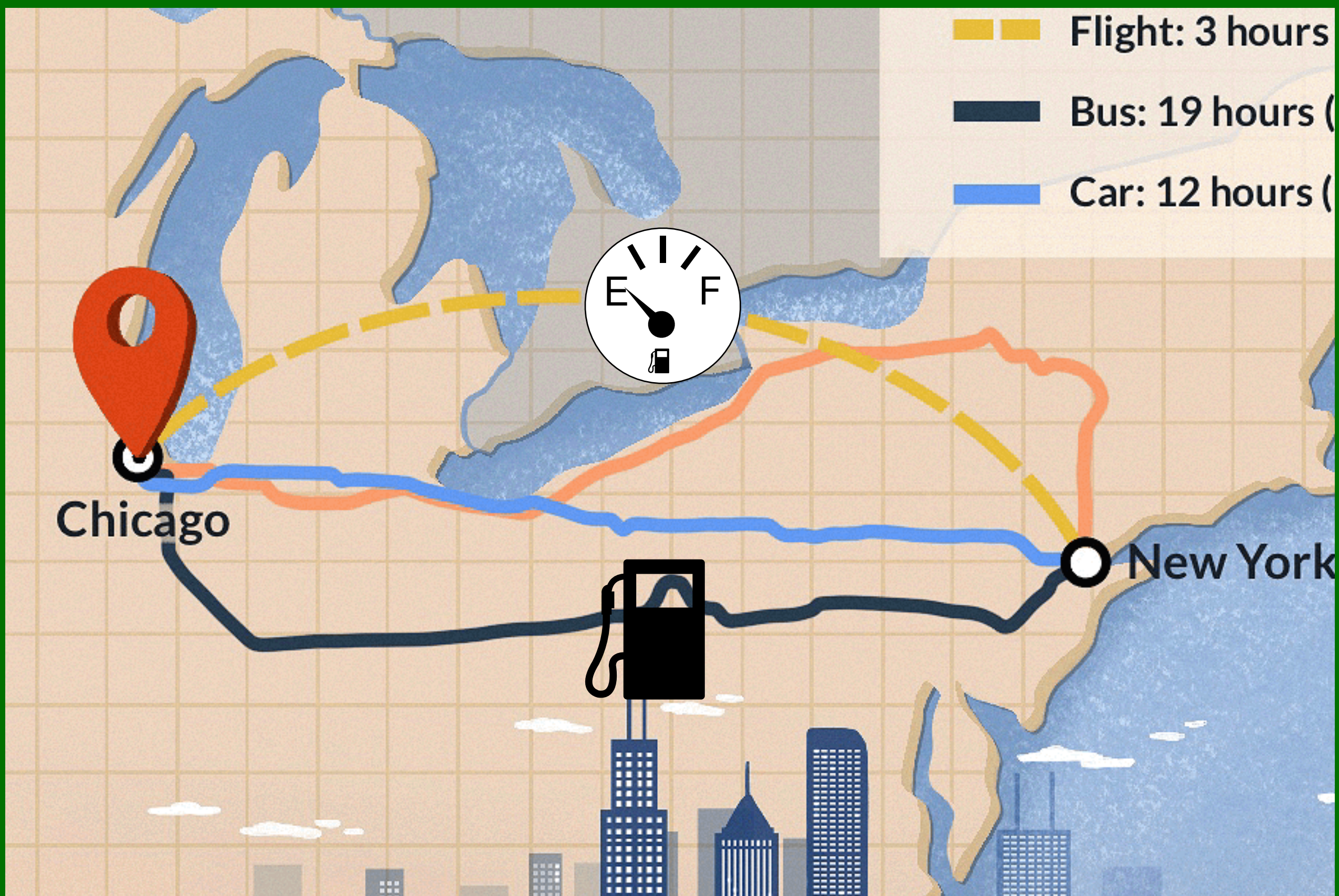
Theorem: For any $\epsilon > 0$, an ϵ -approximate ACE or a certificate that no approximately-feasible policy exists can be computed in polynomial time.

Remark: Our approach relies on the first ever polynomial-time algorithm for solving action-constrained Markov games, which may be of independent interest.

Conclusion

Although exactly compliant policies are much harder to compute in multi-agent settings, approximate policies are not much harder to compute.

Anytime-constrained equilibrium can be approximated in polynomial-time.



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